
Exercises: Advanced Lesson 12

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Your Math Class: MATH 7013 - Advanced Mathematical Modeling

- 1: Evaluate: $(\#^2 \&)$ for the value 3
Evaluate: $\text{Sin}[\#] \&$ for the value $\pi/2$
Evaluate: $\{\#, \text{Sin}[\#]\} \&$ for the value $\pi/2$
Evaluate: $\{\#1, \text{Sin}[\#2]\} \&$ for the value $(1, \pi/2)$
Evaluate: $(\# \geq 10) \&$ for the value 12
Evaluate: $(\# \geq 10) \&$ for the value 8
Evaluate: $(\#1 \geq \#2) \&$ for the value (12, 8)
Evaluate: $(\#1 \geq \#2) \&$ for the value (8, 12)

`Clear[x, f, data, newData];`

`(\#12 &)[3]`

`(Sin[\#1] &)[ $\frac{\pi}{2}$ ]`

`({\#1, Sin[\#1]} &)[ $\frac{\pi}{2}$ ]`

`({\#1, Sin[\#2]} &)[1,  $\frac{\pi}{2}$ ]`

`(\#1 ≥ 10 &)[12]`

`(\#1 ≥ 10 &)[8]`

`(\#1 ≥ \#2 &)[12, 8]`

`(\#1 ≥ \#2 &)[8, 12]`

- 2: Write the commands necessary to select all of the points from the data set:

`data={4, 65, 12, 4, 8, 21, 4};`

that satisfy the criterion of being ≤ 20 using a pure function and the Select command.

`data = {4, 65, 12, 4, 8, 21, 4};`

`Select[data, \#1 ≤ 20 &]`

- 3: Write the commands necessary to select all of the points from the data set:

`data={4, 65, 12, 4, 8, 21, 4};`

that satisfy the criteria of being ≥ 5 and ≤ 20 using a pure function and the Select command.

`Select[data, 5 ≤ \#1 ≤ 20 &]`

4: Write the commands necessary to map a pure function onto the set of data points:

```
data={4, 65, 12, 4, 8, 21, 4};
```

that evaluates the ordered pairs: $\{x, f[x]\}$.

```
Clear[f];
```

```
({#1, f[#1]} &)/@ data
```

5: Write the commands necessary to select all of the points from the data set:

```
newData={{1,4},{3,2},{2,2},{5,1},{2,5}};
```

that satisfy the criterion of having its first coordinate squared being $\leq$ the second coordinate using a pure function and the Select command.

```
newData = {{1, 4}, {3, 2}, {2, 2}, {5, 1}, {2, 5}};
```

```
Select[newData, #1[[1]]2 ≤ #1[[2]] &]
```

6: Evaluate the following lines one line at a time.

```
Clear[f, x]
```

```
f'[x]
```

```
f'
```

```
Sin'[x]
```

```
Sin'
```

```
Clear[f, x];
```

```
f'[x]
```

```
f'
```

```
Sin'[x]
```

```
Sin'
```